Komal Saraf

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EDUCATION:

Bachelor of Technology (B.Tech.) - Computer Science *Sep 2022 - June 2026*
Vellore Institute of Technology (B.TECH CSE)
Cumulative GPA - 7.48

SKILLS:

- **Languages**: Python, C++, SQL.
- **Core Competencies**: Problem-solving, Data Structure and Algorithms, Object-oriented Programming, Database and Management System, (SDLC, STLC, test cases, bug lifecycle management, Process Optimization, Risk Analysis, Machine Learning (TensorFlow, Scikit, Matplotlib, PyTorch, Pandas, NumPy), Trend Prediction, Performance Metrics, Data Analysis.
- **Tools**: Microsoft Excel, Microsoft PowerPoint, Google Sheets, PowerBI, Postman, Jira, MySQL,
- **Soft skills**: analytical, effective verbal and written communication, leadership, collaborative

PROJECTS:

- **CivicHelp AI** *Dec 2025 – Feb 2026*
  - Conceptualized a multimodal GenAI pipeline to automate civic grievance redressal, resulting in a **60% reduction in** manual triage time**.**
  - Managed the full development lifecycle from data ingestion (10K+ records) to deployment, prioritizing features based on municipal urgency and impact.
  - Implemented RAG-based policy grounding using municipal embeddings to ensure factually aligned outputs, reducing misinformation risk and improving action recommendation reliability by 30%.
  - Predicted rising civic trends and urgency via integrated detection tools, streamlining analysis and improving response efficiency by 35%.
  - Tech Stack: Python, Generative AI, RAG, Scikit-learn, TensorFlow, Pandas, NumPy, Matplotlib.

- **Heart disease detection** *May 2025- Sept 2025*
  - Developed a predictive analytics system for early detection of heart disease by training a dataset of patient's physical and medical characteristics.
  - Integrated and optimized various machine learning algorithms (e.g., Logistic Regression, Support Vector Machines, Random Forest) to enhance prediction accuracy and efficiency.
  - Attained a prediction accuracy of more than 85% by analyzing 1000+ dataset entries and built visual reports and performance dashboards to communicate insights
  - Skills Used: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Machine Learning.

- **Desktop Assistant** *Nov 2024- Feb 2025*
  - Developed an automated workflow management system to handle routine operational tasks, resulting in a 20% increase in process efficiency.
  - Designed modular workflows to integrate speech recognition for sub-second voice-to-text conversion and a GUI (Tkinter) for task execution features demonstrating hands-on experience in automation, process streamlining, and user-centric design.
  - Skills used: Automation, OpenCV, PyTorch, Tkinter Python libraries (SpeechRecognition, pyttsx3)

CO-CURRICULARS:

- Python gold badge , Hackerrank.
- C++ gold badge , Hackerrank.
- Member, AI Club, VIT Bhopal 2022 – Coordinated events and collaborated with cross-functional teams for smooth execution, handled planning, communication, and on-ground operations, improved engagement through structured event execution and teamwork, enhancing student participation in Artificial Intelligence initiatives.